GPU Optimization & Resource Report

Face Detection + Blurring Pipeline | Project: AI Research / Real-time Video Privacy

1. Device Settings

Setting	Value
Host CPU	Intel Core i7-14700K (28 logical cores, 31 GB RAM)
GPU	NVIDIA GeForce RTX 5080 (16 GB VRAM)
Driver / CUDA	595.99.02 | CUDA 12.9
Runtime	DeepStream + TensorRT (FP16)
Model	RetinaFace-R50 | input 1280 × 736 | conf 0.15 / NMS 0.4
Source	Live RTSP stream
Frames processed	5 346 (30 warm-up frames excluded from timings)
Run duration	178.7 s

2. Performance Summary

Metric	Value
Avg Pipeline FPS (steady-state)	30.0
Whole-run FPS	30.1
Estimated Max FPS	390.3
Headroom	13.0×
Frames Processed / Lost	5 346 / 0
Keeping up with stream	YES
End-to-end latency (mean / p95 / max)	7.5 ms / 10.1 ms / 23.0 ms

Pipeline is source-limited at ~30 FPS. Estimated max capacity 390 FPS when measured with a file source.

3. Important Modules – GPU vs CPU

Only the modules that matter for understanding where the work happens are listed.

Module	Device
Video Decode (NVDEC)	GPU
Face Detection (TensorRT / RetinaFace-R50)	GPU
Post-process (ROI / NMS / box decode kernels)	GPU
Face Blur (custom CUDA kernels)	GPU
Video Encode (NVENC)	GPU
JPEG encode (nvJPEG)	GPU
Stream Mux / Queues (DeepStream)	GPU
Live preview (D2H + resize + draw)	CPU
Detection & audit log writing	CPU
Probe callbacks & Python glue	CPU
JPEG file write (background thread)	CPU

Almost the entire compute path runs on GPU. CPU handles lightweight bookkeeping, logging and the optional live preview.

4. Resource Usage – Before / During / After Pipeline

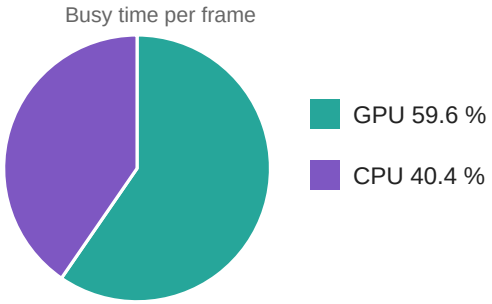
Samples taken at start (init + warm-up), steady state (main run), and at finish.

Metric	At Start	During (Steady)	At Finish
Process CPU (cores)	0.87	0.27	0.26

System CPU (all cores %)	5 %	2 %	3 %
GPU utilisation avg	0 %	6 %	5 %
GPU utilisation peak	—	8 %	—
GPU memory used	3 773 MB	4 364 MB	3 819 MB
GPU power	31 W	74 W	74 W
GPU temperature	34 °C	39 °C	42 °C
Process RAM (RSS)	849 MB	1 266 MB	1 097 MB

5. Workload Split (GPU vs CPU)

Measured busy time per frame in steady state = 6.31 ms (GPU 3.76 ms + CPU 2.55 ms). Host waiting for GPU is only 0.07 ms and is counted separately.



GPU 59.6 % of the busy work · CPU 40.4 % of the busy work.

Of the GPU share, TensorRT inference + NVENC account for the majority; the remaining GPU time is custom CUDA kernels (blur, NMS, ROI) and nvJPEG.

With live preview disabled, the workload shifts further toward the GPU (approximately GPU 70 % / CPU 30 %), as the largest CPU cost item is removed.

6. Top Cost Items (ms per frame)

Stage	Device	ms / frame
TensorRT inference (nvinfer)	GPU	1.94
Live preview (D2H + draw)	CPU	1.15
Kernel launch / enqueue overhead	CPU	1.02
NVENC video encode	GPU	0.82
ROI build + compaction (CuPy)	GPU	0.50

Summary

RetinaFace-R50 pipeline on live RTSP source ran at **30.0 FPS** (source-limited). 0 frames lost. Pipeline headroom **13×** (estimated max ≈ 390 FPS). GPU average utilisation only 6 %; process CPU ≈ 0.27 cores. Busy-time workload split: **GPU 59.6 % / CPU 40.4 %**. Almost all heavy lifting (decode, detection, blur, encode) stays on the GPU; CPU is used mainly for logging, preview and light Python orchestration.